[English Version] Curriculum Vitae - Minwoo Lee

Minwoo Lee

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Portfolio: minuum-portfolio | Research Page: minuum.github.io/MoNaVLA

EDUCATION

Kangnam University — Yongin, South Korea
Bachelor of Science in Artificial Intelligence (Senior)

- Expected Graduation: Feb. 2027
- GPA: **4.27 / 4.5** (Ranked top of the department)

RESEARCH INTERESTS

- Vision-Language-Action (VLA) Models for Robotics
- Embodied AI & Mobile Robot Navigation

- Sim-to-Real Transfer & Physical Latency Mitigation
- Deep Reinforcement Learning

PUBLICATIONS & ACADEMIC CONFERENCES

- **Designing Conversational AI Services Using RAG-Based Loneliness Analysis for Koreans**
  Journal of the Korea Convergence Society (JCCT), KCI Registered, 2024.
  Minwoo Lee (2nd Author), et al.
  - Presented at the International Conference on IPACT 2024 — **Outstanding Paper Award**

RESEARCH EXPERIENCE

Undergraduate Research Assistant — Advised by Prof. Inyeop Choe, Kangnam University
Project: MoNaVLA (Mobile Navigation Vision-Language-Action Models) | *Jan. 2025 – Present*

- **Attention & Failure Analysis:** Diagnosed the VLA backbone's text-path dependency degradation by measuring per-layer attention weights and training a frozen linear probe (val_acc: 96.6%). Confirmed via masking ablation that action predictions reverse when the target object is masked.
- **Architecture Design:** Integrated PaliGemma2 as a zero-shot visual grounder to extract real-time target object position, combining it with CLIP + BBox + L2-norm into a Decomposition structure.
- **Performance:** Achieved a **96.6% closed-loop navigation success rate** (FPE: 0.102m), significantly outperforming the End-to-End baseline (0% success rate).
- **Stepwise Policy Optimization:** Progressively improved MLP navigation policy success rates: **68.4% (Step 1) → 75.9% (Step 2) → 77.0% (Step 3)**.
- **Real-Robot Deployment:** Developed a 10Hz asynchronous controller and Jitter Hold filtering pipeline to bridge the sim-to-real gap on Jetson Orin + ROS2.

- **Award:** Won the **Silver Prize at the 2026-1 Kangnam University Capstone Design Competition**.

PROJECT EXPERIENCE

Serbot 2 Robot Platform Integration — Individual Project | *Jan. 2025 – Mar. 2025*

- Configured a Jetson Orin-based Linux environment and integrated ROS2 driver nodes for the Serbot 2 omnidirectional mobile robot platform.
- Implemented core locomotion control scripts and configured the hardware-software communication interface to host VLA inference models.
- Resolved physical-to-virtual command delays and stabilized actuator control signals.

Ladi — Software Academic Festival | *Oct. 2024*

- Developed an AI-based healthcare routine recommendation system (**Ladi**), winning the **First Prize**.
- Implemented a hybrid search system combining dense (FAISS) and sparse (KiwiBM25) retrievers to optimize medical QA search.
- Integrated a CrossEncoder Reranker to improve recommendation quality, and optimized prompt templates using LangSmith.
- Preprocessed AIHub healthcare QA data to categorize diseases and route intents.

Liberty — AI Academic Festival | *May 2024*

- Served as Project Manager (PM) and lead architect for an AI Legal QA Agent system (**Liberty**), winning the **Excellence Award**.
- Built a multi-agent workflow using LangGraph, integrating dynamic query rewriting and quick filtering.
- Implemented a hybrid retriever (FAISS + KiwiBM25 + Pinecone) achieving 1.000 score in faithfulness and retrieval accuracy benchmarks.

AWARDS & HONORS

- **Silver Prize**, 2026-1 Kangnam University Capstone Design Competition

- **Grand Prize (Dean of College of Engineering Award)**, 2025 Kangnam University Hackathon 'Kangnengthon' (Jointly hosted by Kangnam University & GDGoC: Kangnam University)
- **Outstanding Paper Award**, IPACT 2024 International Conference
- **First Prize (Best Project Award)**, Software Academic Festival (Kangnam University, 2024)
- **Excellence Award**, AI Academic Festival (Kangnam University, 2024)
- **Award**, K-HTML Hackathon (2024)

TECHNICAL SKILLS

- **AI/ML:** PyTorch, HuggingFace, LoRA/PEFT, PaliGemma, CLIP, SigLIP, Kosmos-2, LangGraph (Agent), LangSmith, FAISS, Pinecone (Vector DB), BM25, CrossEncoder (Reranker)
- **Robotics & Infra:** ROS2, Jetson Orin, Linux (Ubuntu), Locomotion Control, Asynchronous Control, Git
- **Programming Languages:** Python, C++, SQL
- **Languages:** Korean (Native), English (Intermediate — preparing for TOEIC)

COURSEWORK

- **Deep Reinforcement Learning** — *Self-study (UC Berkeley CS285: Deep RL course lectures & homework)*
    - Focused on Imitation Learning, Policy Gradients, Actor-Critic methods, and value-based RL.